SWEEP rule on the KD-Toric code

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1 SWEEP Rule Decoder on the Toric

The decoder works as follows: each vertex v looks for a suggestion $\hat{v}$ on $\uparrow(v) \cap lk_2$ that matches σ_v^+. If there are more than one, then pick the one which has the minimal weight.

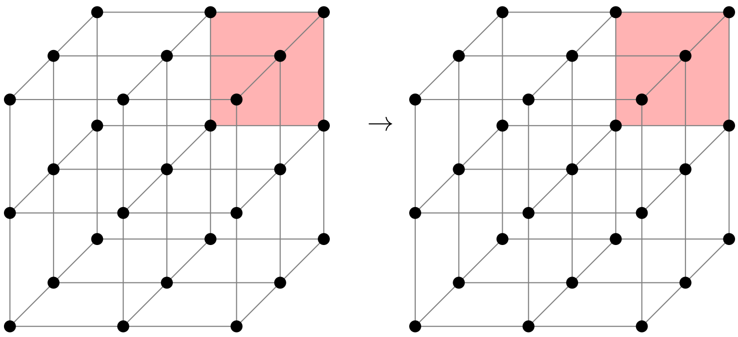

Figure 1: SWEEP rule

2 Span Expansion

Define the potentials functions $L_i : \mathbb{R}^d \to \mathbb{R}$ as follow[1]:

$$L_i(\mathbf{x}) = \begin{cases} \mathbf{x}_i & i < d+1 \\ -\langle \mathbf{1}, \mathbf{x} \rangle & i = d+1 \end{cases}$$

The **Size** of a subset S is defined as:

$$\mathbf{Size}(S) = \sum_i \max_{\mathbf{x} \in S} L_i(\mathbf{x})$$

Numerically, we show that the span property holds for the checks. Namely, let e be an unsatisfied check at time t, and let H_e be the unsatisfied checks at time $t-1$. Then there is a set $H'_e \subset H_e$ such that:

1. For any $i < d+1$ there exists $e' \in H'_e$ such $L_i(e') \geq L_i(e)$.
2. There exists $e' \in H'_e$ such that $L_{d+1}(e') \leq L_{d+1}(e) + 1$.

Proof. Consider a vertex v which at time t adjoins to an unsatisfied edge e, let u, w be vertices which have a positive faces that adjoin to e.

[1]Notice that we use opposite signs relative to Berman and Peter Gacs, yet we agree with Perskil.

1. If $\hat{u}|_e \neq \emptyset$, then the parallel edge to e going out from u belongs to H_e. Denote it by e_u, taking e_u satisfies at least two (three in 4D) of the first d inequalities and the last $d+1$th inequality. Denote by $x \in [d]$ the index of the inequality which isn't satisfied by e_u. So it remains to prove the existence of $e' \in H_e$ for which $L_x(e') \geq L_x(e)$.

   Consider the assignment of bits at time $t-1$ over the complex, and consider the graph $G = (\mathcal{F}, E)$ induced by associating vertices with any face that is set to 1 by the assignment. Two vertices are connected if their origin faces share an edge. Let $K \subset \mathcal{F}$ be a connected component of faces that contains the faces supported on u.

   We say that K is d far from u if:

   $$\max_{f \in K} |f_x - u_x| \leq d$$

   Now, we will prove by induction on d that there is a face $f \in K$ with $f_x \geq v_x$ such one of its edge is not satisfied.

   (a) **Base.** The edge from v which point at the $\hat{z}$ direction must not be satisfied.

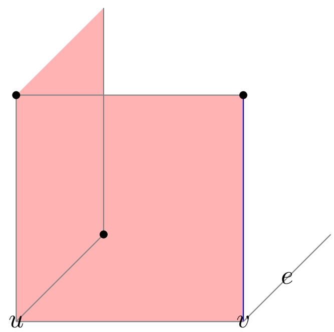

Figure 2: The induction base: In red, we have the faces on the support of u. The blue edge has the same x-coordinate as e. Any attempt to zero out the syndrome on the edge results in adding a face that is at least 2-far from u. Any configuration which is at most 1-far from u has to have that local view, yet might be connected to more faces which don't adjoin the blue edge.

   (b) **Assumption.** Assume that for any K which is at most $d-1$ far from u there is a face, with adjoins edge that is both has a x-coordinate that is greater than u_x and is not satisfied.

   (c) **Induction Step.** Consider a connected component K that is d-far from u. Notice that if by substitute a co-boundary (stabilizer) from K we get K' that is at most d' far from u for some $d' < d$ then we done. That because: $\partial_2 K' = \partial_2(K+z) = \partial_2 K$ for some $z \in \ker \partial_2$.

   On the other hand, **[COMMENT]** Finish that.

2. Else, if for any $u \in \downarrow(u) \cap lk_1$ it holds that $\hat{u}|_e = \emptyset$, then it must be that $e \in H_e$ because $\hat{v}$ can only decrease the syndrome on v. Moreover, it follows that $\hat{v} = \emptyset$ (otherwise, the suggestion would zero out the syndrome on $\uparrow(v) \cap lk_2$). Thus, $\sigma_v \not\subset \uparrow(v) \cap lk_2 \Rightarrow$ there is also an edge $e' \in H_e$ where $e' \in \downarrow(v)$. In that case, take $e \in H_e$ as the edge satisfies the first d equalities and e' for satisfying the $d+1$th inequality.

$\square$

3 Investigation

[COMMENT] To do: understands how the investigation should look like. Plot an example to such tree.
The model goes as follow, at each even iteration we toss a noise, then in each odd iteration we perform a single SWEEP rule iteration. That allows to track over the history of the errors.

We say that a check $e \in$ Error, if there is a face f that adjoin to e and belongs to the Error.

Definition 3.1 (Investigation)**.** For any unsatisfied check e at time t, we define V (the set of checks which participated in 'misleading' of e), and two sets of edges on V, Arrows and Forks via the following algorithm:

let $V = \{e\}$, Arrows $= \emptyset$, Forks $= \emptyset$
let $Q \leftarrow$ queue $\{e\}$
while *Q is not empty* **do**
    $e' \leftarrow$ dequeue Q
    if *$e' \notin$ Error* **then**
        $e_1, .., e_l \leftarrow$ Excuse(e')
        Insert $\{e_i, e'\}$ to Arrows
        Insert $\{e_i, e_j\}$ to Forks
    end
end

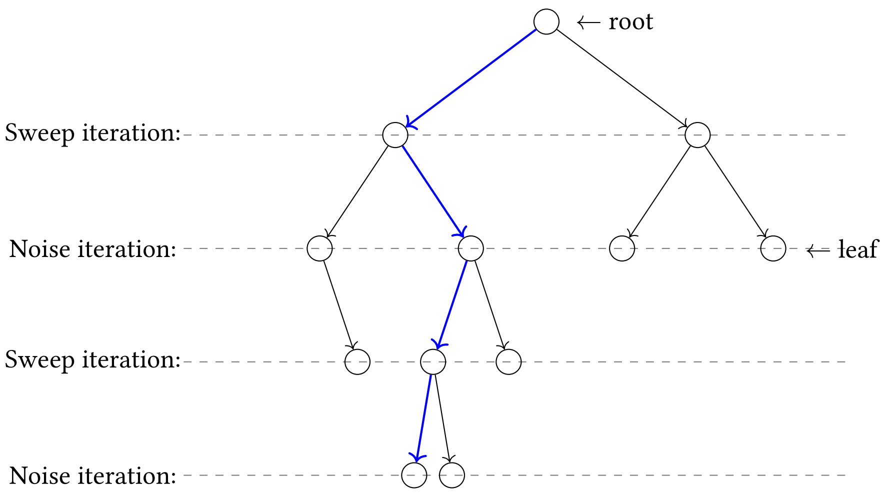

Figure 3: blabla

Definition 3.2 (History, slice.)**.** G_u is the subgraph of $\langle V, \text{Arrows}\rangle$ induced by $\{v \in V : \mathbf{Time}(v) \leq \mathbf{Time}(u)\}$. $\mathbf{History}(u)$ is the connecting connected component of G_u containing u. A slice is an equivalence class of the relation $u \equiv v$ iff $\mathbf{History}(u) = \mathbf{History}(v)$. Notice that if u and v are in the same slice K, then $\mathbf{Time}(u) = \mathbf{Time}(v)$. Thus we can write $\mathbf{Time}(K)$ and $\mathbf{History}(K)$ without ambiguity.

Claim 3.1. If for two different slices K_1 and K_2 we have that $\mathbf{Time}(K_1)) = \mathbf{Time}(K_2))$, then $\mathbf{History}(K_1))$ and $\mathbf{History}(K_2))$ are disjoint.

Proof. Directly from definitions. □

Definition 3.3 (The graph of slice)**.** For a slice K we define the graph $G_K = \langle V_K, E_K\rangle$ as follows:

1. $V_K = \{\text{slice } R : R \subset \mathbf{History}(K) \text{ and } \mathbf{Time}(R) = \mathbf{Time}(K) - 1\}$
2. $\{R, S\} \in E_K$ iff $R, S \in V_K$ and for some $\{v, w\} \in$ Forks we have $v \in R$ and $w \in S$.

Claim 3.2. G_K is connected.

Proof. Assume R and S are from V_K and that $v \in R, w \in S$. Since $v, w \in \mathbf{History}(K)$, there exists a path from v to w in $\mathbf{History}(K)$ consisting of arrows. Select this path in such a way that it contains the minimal number of points x such that $\mathbf{Time}(x) = \mathbf{Time}(K)$. By induction on the number k of such points we prove that R and S are in the same connected component of G_K.

1. Base. $k = 0$, In this case, the path from v to w lies within $G_v(= G_w)$. Hence $\mathbf{History}(v) = \mathbf{History}(w)$ which means that the slices R and S are the identical.
2. Induction step. There exists y on the path from v to w such that $\mathbf{Time}(y) = \mathbf{Time}(K)$. Let x be the point which precedes y on the path, and z the point that follows y. Since the values of $\mathbf{Time}$ at the two ends of an arrow differ by 1, and since $\mathbf{Time}(y)$ is minimal, we know that:

(a) $\mathbf{Excuse}(y) = \langle x, z\rangle$ or $\mathbf{Excuse}(y) = \langle z, x\rangle$, thus $\{x, z\}$ is a fork.

(b) The slices of x and z, say T and U, are both members of V_K.

By the induction hypothesis, R and T are in the same connected component of G_K, and so are U and S. On the other hand, $\{T, U\} \in E_K$.

□

Claim 3.3. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^{k} S_i) \leq \sum_{i=1}^{k} \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned}
\mathbf{Size}(R_1 \cap R_2) &= \sum_{i}^{d+1} \max a_{i,1}, a_{i,2} = \sum_{i}^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_{i}^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}$$

□

Claim 3.4. Let $K = \langle K, v_1, ..v_{d+1}\rangle$ be a spanned slice of the investigation such that $K \neq \mathbf{History}(K)$. Then for some k there exist spanned slices $K_0, .., K_k$ and Forks $F_1, .., F_k$ such that:

1. $\mathbf{Excuse}_i(v_i)$ belongs to poles of $K_0, .., K_k$.

2. Introduce an edge between K_i and K_j iff one of the Forks $F_1, .., F_k$ consists of a pole of K_i and a pole of K_j. Then the graph formed from $K_0, .., K_k$ is connected.

3. $\sum_{i=0}^{k} \mathbf{Span}(K_i) + \sum_{i=1}^{k} \mathbf{Span}(F_i) \geq \mathbf{Span}(K) + 1$

Proof. First Notice that K doesn't contain points for which **Error** is true: they would not connected to any other points in $\mathbf{History}(K)$ via arrows. Thus $\mathbf{Excuse}_i(v_i)$ is defined for $i = 1, .., d+1$. Next consider the graph $G_K = \langle V_K, E_K\rangle$ from **??**. V_K consists of those R for which $\mathbf{Time}(R) + 1 = \mathbf{Time}(K)$ and which are contained in $\mathbf{History}(K)$. Since $\{v_i, \mathbf{Excuse}_i(v_i)\}$ is an arrow $\mathbf{Excuse}_i(v_i)$ belongs to a slice R_i from V_K, for $i = 1, .., d+1$.

Since G_K is connected and its edges are realized by forks, there exist a minimal collection $\{K_0, .., K_k\}$ of slices form V_K which contains $R_1, R_2, ..R_{d+1}$, and which is connected in G_K. We pick a set of forks $\mathbf{F} = \{F_1, .., F_k\}$ which realizes a spanning tree for this collection of slices. Later we will identify edges from this spanning tree with the forks from $\mathbf{F}$.

For $i = 1, .., d+1$ we set $\mathbf{Excuse}_i(v_i)$ to be the ith pole of R_i, This assures (1). The set of remaining poles for $K_0, .., K_k$ will be equal to $\cup_{i=k}^{k} F_i$. This will assure (ii), We will also select poles for $F_1, .., F_k$ to form 'spanned forks' $F_1, \cdots, F_k$ in such way that:

$$\sum_{i=0}^{k} \mathbf{Span}(K_i) + \sum_{i=1}^{k} \mathbf{Span}(F_i) = \sum_{i=1}^{d+1} L_i\left(\mathbf{Excuse}_i(v_i)\right) \geq \mathbf{Span}(K) + 1$$

This will assure (3).

Now we describe the selection of poles. Since we have to span.. □

4 Explantation of a Spanned Slice

[COMMENT] Here we follow the original proof, but change the triminalogy, m will be the number of leaves in the explanation, then in the end we conclude that the number of faults is at least m/Δ. So $|R| \leq f(\frac{m}{\Delta})$.

Definition 4.1. Given sets W of points, R of arrows and F of forks, we define U as the set of vertices of the graph induced by the edges of $R \cup F$. The sets W, R and F form an explanation of a spanned slice $K = \langle K, v_1, .., v_{d+1} \rangle$ iff:

1. $W \cup \{v_1, .., v_{d+1}\} \subset U \subset \mathbf{History}(K)$ and the graph $\langle U, R \cup F \rangle$ is connected.
2. All elements of W are errors.
3. For $m = |W|$ and $s = \mathbf{Span}(K)$ we have $|R| \leq \alpha(m-1) - \beta s$ and $|F| < m - 1$.

Claim 4.1. Every spanned slice $K = \langle K, v_1, .., v_{d+1} \rangle$ has an explanation.

Proof. Let $h = \mathbf{Time}(K) - \min_{u \in \mathbf{History}(K)} \mathbf{Time}(u)$. The proof is by induction on h.

1. Base. $h = 0$. Then no arrows can connect points of $\mathbf{History}(K)$, Thus $\mathbf{History}(K)$, as well as $\mathbf{K}$ consists of a single point u such that $u \in$ Error **[COMMENT]** Leaves. In that case $U = W = \{u\}$, $m = 1, s = 0$ and $|R|, |F| = 0$ satisfies the required.
2. Induction Step. Now, we know that $K \neq \mathbf{History}(K)$. Thus, by **??**, we know that for some k there exist spanned slices $K_0, .., K_k$ and forks $F_1, .., F_k$ that satisfy **??**.

   Let $s_i = \mathbf{Span}(K_i)$. By the inductive hypothesis, we know that K_i has an explanation $\mathbf{W}_i, \mathbf{R}_i, \mathbf{F}_i$. For $m_i = |\mathbf{W}_i|$, the sets R_i and F_i have at most $\alpha m_i - \beta s_i$ and $m_i - 1$ leaves. We define:

   (a) $\mathbf{W} = \cup_{i=0}^{k} \mathbf{W}_i$, with $m = \sum_{i=0}^{k} m_i$ leaves. Notice that $\mathbf{W}_i \subset K_i$ and that $\mathbf{Time}(K_i) = \mathbf{Time}(K_j)$ for all i, j, thus by **??** $\mathbf{W}$ is a union of disjoint set.

   (b) $\mathbf{F} = \{F_1, .., F_k\} \cup_{i=0}^{k} \mathbf{F}_i$ with at most $k + \sum_{i=0}^{k} m_i - 1 = m - 1$ elements.

   (c) $\mathbf{R} = \{\{v_i, \mathbf{Excuse}_i(v_i)\}\} \cup_{i=0}^{k} \mathbf{R}_i$ with at most

   $$\begin{aligned} d + 1 + \sum_{i=0}^{k} \alpha(m_i - 1) - \beta s_i &\leq d + 1 + \alpha m - \alpha(k+1) - \beta\left(s - 1 + k \cdot \mathbf{Span}(\text{fork})\right) \\ =& \alpha m - \beta s + (d + 1 - (k+1)\alpha + \beta - \beta k \cdot \mathbf{Span}(\text{fork})) \end{aligned}$$

   So its enough to pick α, β such that for any $k \geq 1$:

   $$\begin{aligned} & d + 1 - (k+1)\alpha + \beta - \beta k f \leq 0 \\ \Rightarrow & \beta = 1, \alpha = (d+1)^3(1+f) \end{aligned}$$

□